



EcoSort

Your waste, in the right place.

Every day, a seconds-long decision can contaminate an entire recyclable batch.

Multimodal AI

Photo + text

Immediate action

Dario Pomasqui | Yachay Tech University | Neural Networks and Deep Learning

What Problem Does It Solve?

2 min

The doubt appears in your hand

“I have a juice box”

“These batteries no longer work”

“Leftover food from yesterday”

Users do not need theory;
they need a clear decision.

Who is affected



Households

Daily sorting without long guides



Campus

Mixed waste in classrooms and cafeterias



Cities

Contaminated recycling and special routes

When waste is sorted incorrectly,
environmental value is lost.

The Solution

2 min



EcoSort does not only say what it is.

It tells you what to do with it.

1. The user takes a photo

2. Writes a short description

3. Gets a verdict, steps, and alerts

recycle

clean

donate

compost

special point

A real use case

- 1 **Photo: AA batteries**
Result: battery
- 2 **Text: “from an old remote”**
Action: special point
- 3 **Photo: dinner leftovers**
Result: organic waste
- 4 **Text: “from yesterday”**
Action: organics or compost

Live demo

EcoSort

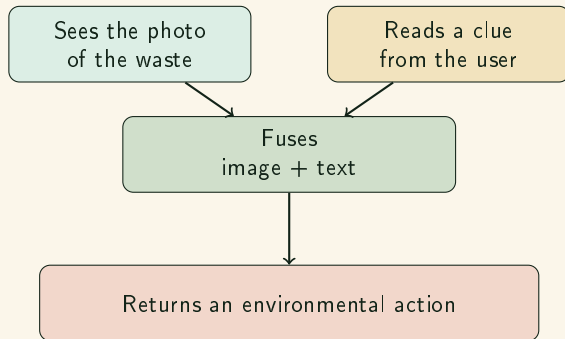
Scan
Describe
Act

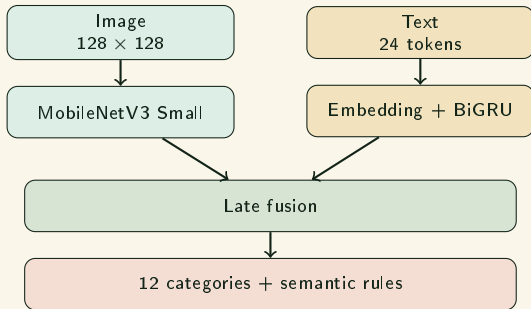
The demo can be live in the app or shown as a short video recorded from the phone.

No code needed

The neural network recognizes the material in the image.

The semantic layer reads context: clean, chemical, sanitary, or reusable.





Architectural decisions

MobileNetV3 lightweight mobile vision

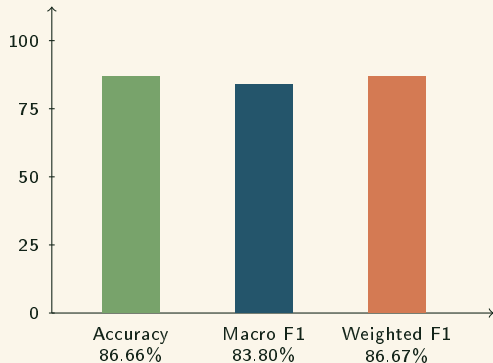
BiGRU short descriptions

Late fusion explainable decision

Semantics label to action

Results and Validation

1-2 min



Functional validation

13/13

natural scenarios
classified correctly

Without writing: “to recycle”,
“for compost” or “special point”.

The app infers the recommendation.

What We Learned

1 min



**Vision alone
is not enough**

Context changes
the final decision.



**AI + rules
is more useful**

Not just a label:
it turns into action.



**Users
need clarity**

Short phrases,
concrete steps,
simple alerts.

EcoSort turns confusion into environmental action

Product

Photo + text
+ plan

Impact

Less
contamination

Innovation

Multimodal AI
+ semantics

Scale

Homes, campus
and schools

**A simple app for real users, built around
a multimodal neural network.**



EcoSort

It is not only about throwing away less.
It is about deciding better.

Your waste, in the right place.

Dario Pomasqui | Yachay Tech University